

Ex:No: 6A	Customer Purchase Category Prediction Using Naive Bayes Classifier
Date:	

Problem Description:

An e-commerce company wants to predict whether a customer will make a **High** or **Low** purchase based on demographic and behavioral attributes. The dataset contains both **categorical** (Gender, Membership) and **numerical** (Age, Visits per Month) features.

To perform probabilistic classification and handle mixed data types efficiently, the **Naive Bayes classifier** is used.

Objective:

To implement the Naive Bayes classification algorithm in R by training a model on customer data, predicting purchase categories, and evaluating classification accuracy.

```

1 customer_data<-data.frame(
2   Age = c(22, 35, 40, 28, 50, 45, 30, 55, 27, 38),
3   Visits = c(2, 5, 6, 3, 8, 7, 4, 9, 3, 6),
4   Gender = c("Male", "Female", "Female", "Male", "Female", "Male", "Female", "Male", "Male", "Female"),
5   Membership = c("Regular", "Gold", "Gold", "Regular", "Gold", "Silver", "Silver", "Gold", "Silver", "Gold"),
6   Purchase = c("Low", "High", "High", "Low", "High", "High", "Medium", "High", "Low", "High")
7 )
8 customer_data
9 customer_data$Gender<-as.factor(customer_data$Gender)
10 customer_data$Membership<-as.factor(customer_data$Membership)
11 customer_data$Purchase<-as.factor(customer_data$Purchase)
12 index<-sample(1:nrow(customer_data), 0.7*nrow(customer_data))
13 train_data<-customer_data[index,]
14 test_data<-customer_data[-index,]
15 library(e1071)
16 nb_model<-naiveBayes(Purchase~.,data = train_data)
17 nb_model
18 predicted_purchase<-predict(nb_model, test_data)
19 predicted_purchase
20 confusion_matrix<-table(
21   Actual = test_data$Purchase,
22   Predicted = predicted_purchase
23 )
24 confusion_matrix
25 accuracy = sum(diag(confusion_matrix))/sum(confusion_matrix)
26 accuracy
27 barplot(table(customer_data$Purchase),
28   col = c("navyblue", "brown", "darkgreen"),
29   main = "Purchase Category Distribution",
30   xlab= "Purchase Category",
31   ylab = "Number of Customers"
32 )
33 plot(customer_data$Age, customer_data$Visits,
34   col = as.numeric(customer_data$Purchase),
35   pch = 19,
36   main = "Customer Distribution By Purchase(Category)",
37   xlab = "Age",
38   ylab = "Monthly Visits"
39 )

```

```

)
legend("topleft",
      legend = levels(customer_data$Purchase),
      col = 1:length(levels(customer_data$Purchase)),
      pch = 19
)
boxplot(Age~Purchase, data = customer_data,
      col = c("yellow", "blue", "red"),
      main = "Age Distribution by Purchase category",
      xlab = "Purchase Category",
      ylab = "Age"
)
plot(test_data$Age, test_data$Visits,
      col = as.numeric(predicted_purchase),
      pch = 19,
      main = "Naive Bayes Classification Results",
      xlab = "Age",
      ylab = "Monthly Visits"
)
legend("topleft",
      legend = levels(predicted_purchase),
      col = 1:length(levels(predicted_purchase)),
      pch = 19
)

```

```
> customer_data
```

	Age	Visits	Gender	Membership	Purchase
1	22	2	Male	Regular	Low
2	35	5	Female	Gold	High
3	40	6	Female	Gold	High
4	28	3	Male	Regular	Low
5	50	8	Female	Gold	High
6	45	7	Male	Silver	High
7	30	4	Female	Silver	Medium
8	55	9	Male	Gold	High
9	27	3	Male	Regular	Low
10	38	6	Female	Silver	Medium

```
> nb_model

Naive Bayes Classifier for Discrete Predictors

Call:
naiveBayes.default(x = X, y = Y, laplace = laplace)

A-priori probabilities:
Y
      High      Low      Medium
0.5714286 0.2857143 0.1428571

Conditional probabilities:

      Age
Y      [,1]      [,2]
High  43.75  8.539126
Low   24.50  3.535534
Medium 38.00      NA

      visits
Y      [,1]      [,2]
High   6.75  1.7078251
Low    2.50  0.7071068
Medium 6.00      NA

      Gender
Y      Female Male
High   0.5    0.5
Low    0.0    1.0
Medium 1.0    0.0

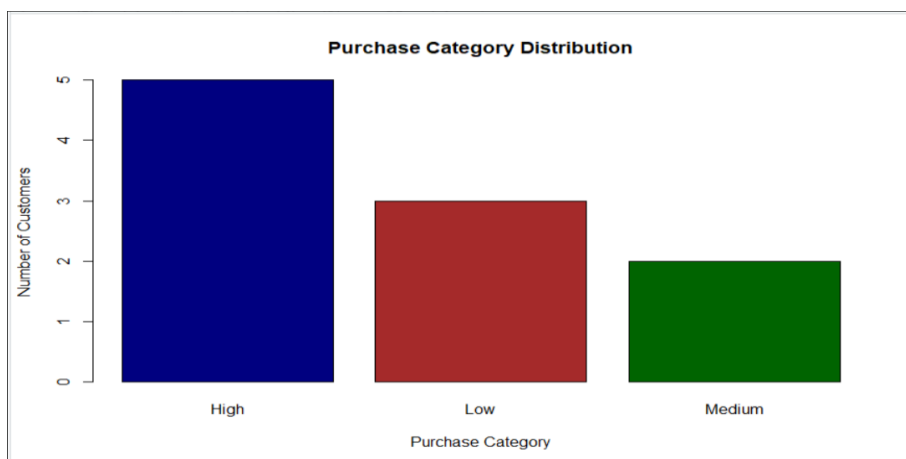
      Membership
Y      Gold Regular Silver
High   0.75    0.00    0.25
Low    0.00    1.00    0.00
Medium 0.00    0.00    1.00

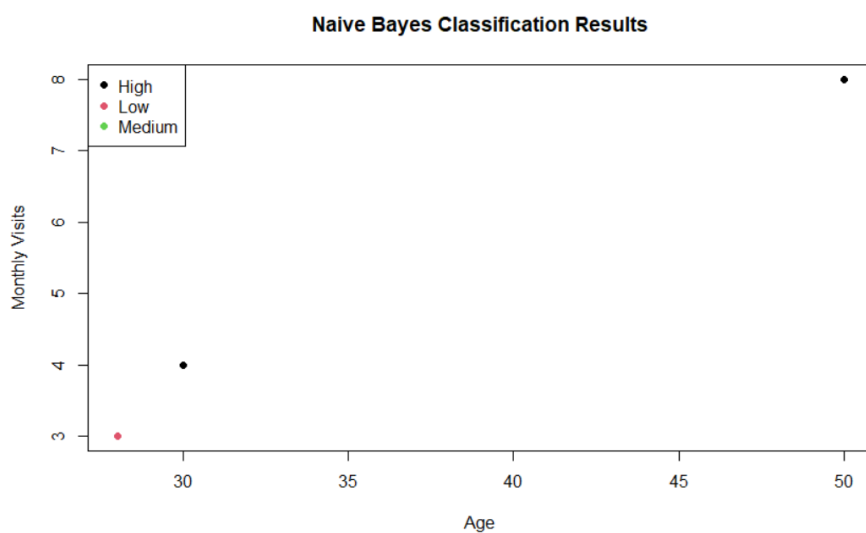
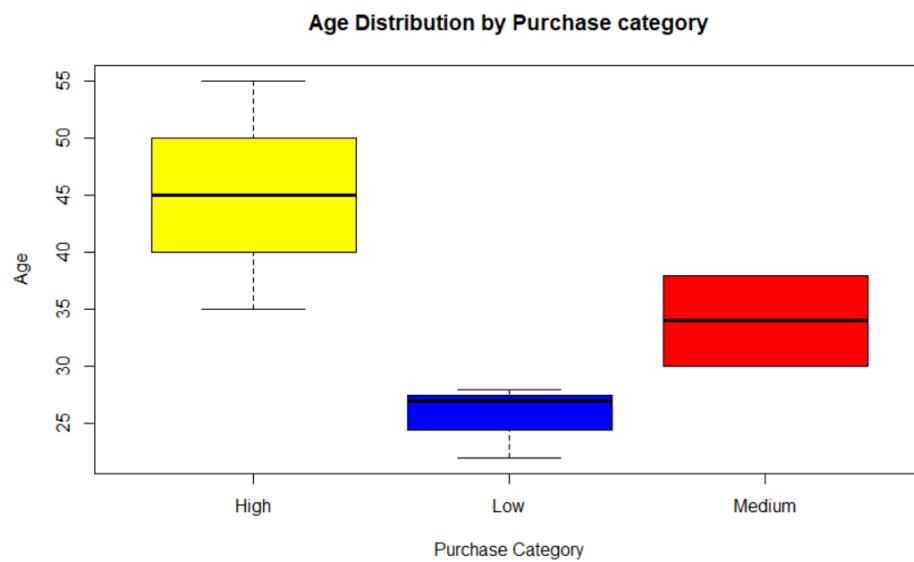
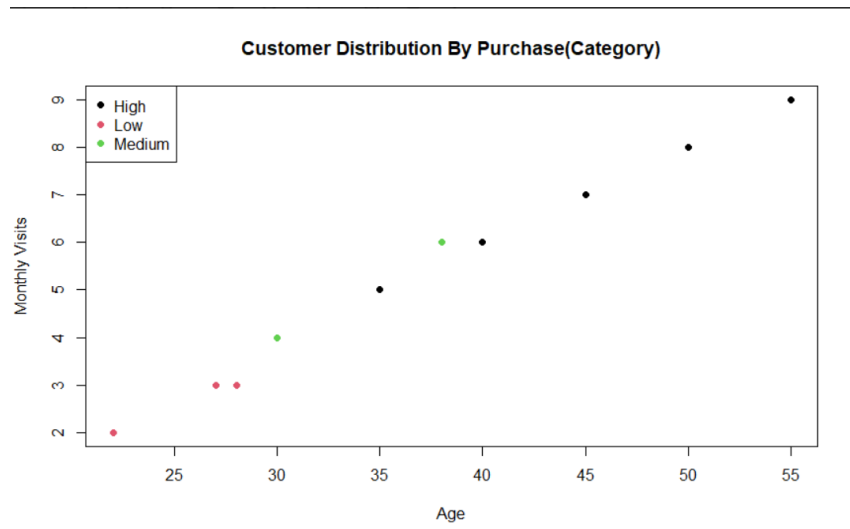
> predicted_purchase<-predict(nb_model, test_data)
> predicted_purchase
[1] Low  High High
Levels: High Low Medium
```

```
> confusion_matrix

      Predicted
Actual  High Low Medium
High    1  0    0
Low     0  1    0
Medium  1  0    0

> accuracy = sum(diag(confusion_matrix))/sum(confusion_matrix)
> accuracy
[1] 0.6666667
```





Ex:No: 6B

Date:

Iris Flower Species Classification Using Naive Bayes Algorithm

Problem Description:

The Iris dataset contains measurements of iris flowers belonging to three species: **Setosa**, **Versicolor**, and **Virginica**. Each flower is described using four numerical attributes: **Sepal Length**, **Sepal Width**, **Petal Length**, and **Petal Width**.

The objective is to classify iris flowers into their respective species using the **Naive Bayes classifier** and visualize the data distribution and classification results.

Objective:

To implement the Naive Bayes classification algorithm in R for predicting iris flower species and to visualize the dataset and classification results.

```
1 library(e1071)
2 library(caret)
3 library(ggplot2)
4 data(iris)
5 head(iris)
6 set.seed(123)
7 library(caret)
8 train_index<-createDataPartition(iris$Species, p=0.7, list=FALSE)
9 train_data<-iris[train_index,]
10 test_data<-iris[-train_index,]
11 library(e1071)
12 nb_model<-naiveBayes(Species~., data = train_data)
13 print(nb_model)
14 library(caret)
15 conf_matrix<-confusionMatrix(predictions, test_data$Species)
16 print(conf_matrix$overall['Accuracy'])
17 predictions<-predict(nb_model, test_data)
18 print(predictions)
19 test_data$Predicted_Species<-predictions
20 ggplot(test_data, aes(x = Sepal.Length, y = Sepal.Width, color = Predicted_Species))+
21   geom_point(size = 3) + ggtitle("Naive Bayes Classification Results")
22
```

```
> nb_model<-naiveBayes(Species~., data = train_data)
> print(nb_model)
```

Naive Bayes Classifier for Discrete Predictors

Call:

```
naiveBayes.default(x = X, y = Y, laplace = laplace)
```

A-priori probabilities:

```
Y
      setosa versicolor virginica
0.3333333 0.3333333 0.3333333
```

Conditional probabilities:

```
      Sepal.Length
Y      [,1]      [,2]
setosa 4.991429 0.3657156
versicolor 5.942857 0.4558674
virginica 6.631429 0.6846087
```

```
      Sepal.Width
Y      [,1]      [,2]
setosa 3.365714 0.3455114
versicolor 2.777143 0.3227508
virginica 2.982857 0.3222035
```

```
      Petal.Length
Y      [,1]      [,2]
setosa 1.471429 0.1808012
versicolor 4.262857 0.4222389
virginica 5.591429 0.5741344
```

```
      Petal.Width
Y      [,1]      [,2]
setosa 0.2314286 0.09321521
versicolor 1.3285714 0.20518489
virginica 2.0342857 0.25775371
```

```
> data(iris)
```

```
> head(iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

```
> predictions<-predict(nb_model, test_data)
```

```
> print(predictions)
```

```
[1] setosa setosa setosa setosa setosa setosa setosa
[8] setosa setosa setosa setosa setosa setosa setosa
[15] setosa versicolor virginica versicolor versicolor versicolor versicolor
[22] virginica versicolor versicolor versicolor versicolor versicolor versicolor
[29] versicolor versicolor virginica virginica virginica virginica virginica
[36] virginica versicolor virginica virginica virginica versicolor virginica
[43] virginica virginica virginica
Levels: setosa versicolor virginica
```

```
> print(conf_matrix$overall['Accuracy'])  
Accuracy  
0.9111111
```

